# BOS : Vectors

# Part A

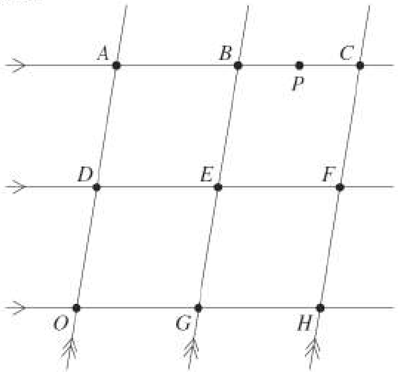
**Extension**

**NAME:**

**2020 Q1**

The diagram shows a grid of equally spaced lines. The vector $\overrightarrow{OH} = \underset{\sim}{h}$ and the vector $\overrightarrow{OA} = \underset{\sim}{a}$. The point $P$ is halfway between $B$ and $C$. **1**

Which expression represents the vector $\overrightarrow{OP}$?




A. $-\frac{1}{2}\underset{\sim}{a} - \frac{1}{4}\underset{\sim}{h}$

B. $\frac{1}{4}\underset{\sim}{a} - \frac{1}{2}\underset{\sim}{h}$

C. $\underset{\sim}{a} + \frac{1}{4}\underset{\sim}{h}$

D. $\underset{\sim}{a} + \frac{3}{4}\underset{\sim}{h}$

**2020 Q2**

What is the angle between the vectors $\begin{pmatrix}7\\1\end{pmatrix}$ and $\begin{pmatrix}-1\\1\end{pmatrix}$? **1**

A. $\cos^{-1}(0.6)$

B. $\cos^{-1}(0.06)$

C. $\cos^{-1}(-0.06)$

D. $\cos^{-1}(-0.6)$

**2020 Q3**

A force described by the vector $\underset{\sim}{F} = \begin{pmatrix}2\\1\end{pmatrix}$ newtons is applied to an object lying on a line $\ell$ which is parallel to the vector $\begin{pmatrix}3\\4\end{pmatrix}$.

(i) Find the component of the force $\underset{\sim}{F}$ in the direction of the line $\ell$. **2**

(ii) What is the component of the force $\underset{\sim}{F}$ in the direction perpendicular to the line? **1**

**(i)** A unit vector in the direction of $\ell$ is $\hat{\underset{\sim}{v}} = \frac{1}{5}\begin{pmatrix}3\\4\end{pmatrix}$, so the component of $\underset{\sim}{F}$ in the direction of $\ell$ is

$$\left(\underset{\sim}{F}\cdot\hat{\underset{\sim}{v}}\right)\hat{\underset{\sim}{v}} = 2\hat{\underset{\sim}{v}} = \begin{pmatrix}1.2\\1.6\end{pmatrix}.$$

**(ii)** The component of $\underset{\sim}{F}$ perpendicular to $\ell$ is $\underset{\sim}{F} - \begin{pmatrix}1.2\\1.6\end{pmatrix} = \begin{pmatrix}0.8\\-0.6\end{pmatrix}$.

**2020 Q4**

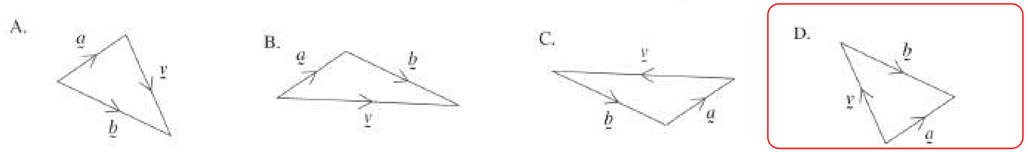
Maria starts at the origin and walks along all of the vector $2\underset{\sim}{i} + 3\underset{\sim}{j}$, then walks along all of the vector $3\underset{\sim}{i} - 2\underset{\sim}{j}$ and finally along all of the vector $4\underset{\sim}{i} - 3\underset{\sim}{j}$. **1**

How far from the origin is she?

A. $\sqrt{77}$ B. $\sqrt{85}$ C. $2\sqrt{13} + \sqrt{5}$ D. $\sqrt{5} + \sqrt{7} + \sqrt{13}$

**2020 Q5**

The vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are shown. Which diagram below shows the vector $\underset{\sim}{v} = \underset{\sim}{a} - \underset{\sim}{b}$? **1**


**2020 Q6**

The projection of the vector $\begin{pmatrix}6\\7\end{pmatrix}$ onto the line $y = 2x$ is $\begin{pmatrix}4\\8\end{pmatrix}$. **1**

The point $(6, 7)$ is reflected in the line $y = 2x$ to a point $A$.

What is the position vector of the point $A$?

A. $\begin{pmatrix}6\\12\end{pmatrix}$ B. $\begin{pmatrix}2\\9\end{pmatrix}$ C. $\begin{pmatrix}-6\\7\end{pmatrix}$ D. $\begin{pmatrix}-2\\1\end{pmatrix}$

**2020 Q7**

For what value(s) of $a$ are the vectors $\begin{pmatrix}a\\-1\end{pmatrix}$ and $\begin{pmatrix}2a-3\\2\end{pmatrix}$ perpendicular? **3**

Since the vectors are perpendicular.

$$0 = \begin{pmatrix}a\\-1\end{pmatrix}\cdot\begin{pmatrix}2a-3\\2\end{pmatrix}$$

$$0 = 2a^2 - 3a - 2$$

$$0 = (2a+1)(a-2)$$

$$a = -\frac{1}{2} \quad \text{or} \quad 2$$

**2021 Q8**

Given that $\overrightarrow{OP} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ and $\overrightarrow{OQ} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, what is $\overrightarrow{PQ}$? **1**

A. $\begin{pmatrix} 1 \\ -6 \end{pmatrix}$ B. $\begin{pmatrix} -1 \\ 6 \end{pmatrix}$ C. $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$ D. $\begin{pmatrix} -5 \\ -4 \end{pmatrix}$

**2021 Q9**

For the two vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ it is known that $\overrightarrow{OA} \cdot \overrightarrow{OB} < 0$. **1**

Which of the following statements MUST be true?

A. Either, $\overrightarrow{OA}$ is negative and $\overrightarrow{OB}$ is positive, or, $\overrightarrow{OA}$ is positive and $\overrightarrow{OB}$ is negative.

B. The angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is obtuse.

C. The product $|\overrightarrow{OA}||\overrightarrow{OB}|$ is negative.

D. The points $O$, $A$ and $B$ are collinear.

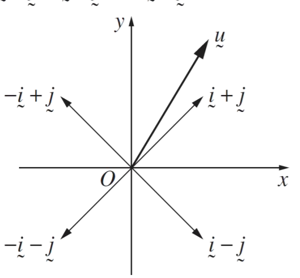
**2021 Q10**

Find $(\underset{\sim}{i} + 6\underset{\sim}{j}) + (2\underset{\sim}{i} - 7\underset{\sim}{j})$. **1**

$$(\underset{\sim}{i} + 6\underset{\sim}{j}) + (2\underset{\sim}{i} - 7\underset{\sim}{j}) = 3\underset{\sim}{i} - \underset{\sim}{j}$$

**2022 Q11**

The following diagram shows the vector $\underset{\sim}{u}$ and the vectors $\underset{\sim}{i} + \underset{\sim}{j}$, $-\underset{\sim}{i} + \underset{\sim}{j}$, $-\underset{\sim}{i} - \underset{\sim}{j}$ and $\underset{\sim}{i} - \underset{\sim}{j}$. **1**




Which statement regarding this diagram could be true?

A. The projection of $\underset{\sim}{u}$ onto $\underset{\sim}{i} + \underset{\sim}{j}$ is the vector $1.1\underset{\sim}{i} + 1.8\underset{\sim}{j}$.

B. The projection of $\underset{\sim}{u}$ onto $-\underset{\sim}{i} + \underset{\sim}{j}$ is the vector $-0.4\underset{\sim}{i} + 0.4\underset{\sim}{j}$.

C. The projection of $\underset{\sim}{u}$ onto $-\underset{\sim}{i} - \underset{\sim}{j}$ is the vector $3.2\underset{\sim}{i} + 3.2\underset{\sim}{j}$.

D. The projection of $\underset{\sim}{u}$ onto $\underset{\sim}{i} - \underset{\sim}{j}$ is the vector $0.5\underset{\sim}{i} - 0.5\underset{\sim}{j}$.

**2022 Q12**

The angle between two unit vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is $\theta$ and $|\underset{\sim}{a} + \underset{\sim}{b}| < 1$. **1**

Which of the following best describes the possible range of values of $\theta$?

A. $0 \le \theta < \frac{\pi}{3}$ B. $0 \le \theta < \frac{2\pi}{3}$ C. $\frac{\pi}{3} < \theta \le \pi$ D. $\frac{2\pi}{3} < \theta \le \pi$

**2022 Q13**

For the vectors $\underset{\sim}{u} = \underset{\sim}{i} - \underset{\sim}{j}$ and $\underset{\sim}{v} = 2\underset{\sim}{i} + \underset{\sim}{j}$, evaluate each of the following.

(i) $\underset{\sim}{u} + 3\underset{\sim}{v}$ (ii) $\underset{\sim}{u} \cdot \underset{\sim}{v}$ **1, 1**

**Sample answer:**

$\underset{\sim}{u} = \underset{\sim}{i} - \underset{\sim}{j}$, $\underset{\sim}{v} = 2\underset{\sim}{i} + \underset{\sim}{j}$

$$\underset{\sim}{u} + 3\underset{\sim}{v} = \underset{\sim}{i} - \underset{\sim}{j} + 3(2\underset{\sim}{i} + \underset{\sim}{j})$$
$$= \underset{\sim}{i} - \underset{\sim}{j} + 6\underset{\sim}{i} + 3\underset{\sim}{j} = 7\underset{\sim}{i} + 2\underset{\sim}{j}$$

**Sample answer:**

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = 1 \times 2 + (-1) \times 1$$
$$= 1$$

**2022 Q14**

The vectors $\underset{\sim}{u} = \begin{pmatrix} a \\ 2 \end{pmatrix}$ and $\underset{\sim}{v} = \begin{pmatrix} a-7 \\ 4a-1 \end{pmatrix}$ are perpendicular. **2**

What are the possible values of $a$?

$\underset{\sim}{u}$ and $\underset{\sim}{v}$ are perpendicular $\therefore \underset{\sim}{u} \cdot \underset{\sim}{v} = 0$

That is $a(a-7) + 2(4a-1) = 0$

$a^2 - 7a + 8a - 2 = 0$

$a^2 + a - 2 = 0$

$(a+2)(a-1) = 0$

$\therefore a = -2$ or $1$

**2020 Q7**

For what value(s) of $a$ are the vectors $\begin{pmatrix} a \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 2a-3 \\ 2 \end{pmatrix}$ perpendicular? **3**

Since the vectors are perpendicular,

$$0 = \begin{pmatrix} a \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2a-3 \\ 2 \end{pmatrix}$$
$$0 = 2a^2 - 3a - 2$$
$$0 = (2a+1)(a-2)$$
$$a = -\frac{1}{2} \text{ or } 2$$

**2020 Q1**

The points $A$ and $B$ are fixed points in a plane and have position vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ respectively.

The point $P$ with position vector $\underset{\sim}{p}$ also lies in the plane and is chosen so that $\angle APB = 90°$.

(i) Explain why $(\underset{\sim}{a} - \underset{\sim}{p}) \cdot (\underset{\sim}{b} - \underset{\sim}{p}) = 0$. **1**

(ii) Let $\underset{\sim}{m} = \frac{1}{2}(\underset{\sim}{a} + \underset{\sim}{b})$ denote the position vector of $M$, the midpoint of $A$ and $B$. **3**

Using the properties of vectors, show that $|\underset{\sim}{p} - \underset{\sim}{m}|^2$ is independent of $\underset{\sim}{p}$ and find its value.

(iii) What does the result in part (ii) prove about the point $P$? **1**

**(i)** $\overrightarrow{PA} = (\underset{\sim}{a} - \underset{\sim}{p})$, while $\overrightarrow{PB} = (\underset{\sim}{b} - \underset{\sim}{p})$. Since we are given that they are perpendicular, the dot product of these two vectors is zero.

**(ii)** From part (i) $\underset{\sim}{a} \cdot \underset{\sim}{b} - \underset{\sim}{p} \cdot (\underset{\sim}{a} + \underset{\sim}{b}) + \underset{\sim}{p} \cdot \underset{\sim}{p} = 0$, so

$$|\underset{\sim}{p} - \underset{\sim}{m}|^2 = \frac{1}{4}(2\underset{\sim}{p} - (\underset{\sim}{a} + \underset{\sim}{b})) \cdot (2\underset{\sim}{p} - (\underset{\sim}{a} + \underset{\sim}{b}))$$

$$= \underset{\sim}{p} \cdot \underset{\sim}{p} - \underset{\sim}{p} \cdot (\underset{\sim}{a} + \underset{\sim}{b}) \cdot \frac{1}{4}(\underset{\sim}{a} + \underset{\sim}{b}) \cdot (\underset{\sim}{a} + \underset{\sim}{b})$$

$$= \frac{1}{4}(\underset{\sim}{a} + \underset{\sim}{b}) \cdot (\underset{\sim}{a} + \underset{\sim}{b}) - \underset{\sim}{a} \cdot \underset{\sim}{b} \qquad \text{independent of } \underset{\sim}{p}$$

**(iii)** $P$ lies on the circle whose diameter is $AB$.

OR

$P$ lies on a circle centre $M$ and radius $\sqrt{\frac{1}{4}(\underset{\sim}{a} + \underset{\sim}{b}) \cdot (\underset{\sim}{a} + \underset{\sim}{b}) - \underset{\sim}{a} \cdot \underset{\sim}{b}}$.

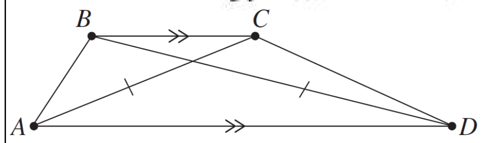
**2021 Q2**

(i) For vector $\underset{\sim}{v}$, show that $\underset{\sim}{v} \cdot \underset{\sim}{v} = |\underset{\sim}{v}|^2$. **1**

(ii) In the trapezium $ABCD$, $BC$ is parallel to $AD$ and $|\overrightarrow{AC}| = |\overrightarrow{BD}|$.

Let $\underset{\sim}{a} = \overrightarrow{AB}$, $\underset{\sim}{b} = \overrightarrow{BC}$ and $\overrightarrow{AD} = k\overrightarrow{BC}$, where $k > 0$. **3**

Using part (i), or otherwise, show $2\underset{\sim}{a} \cdot \underset{\sim}{b} + (1-k)|\underset{\sim}{b}|^2 = 0$.




***Sample answer:***

Let $\underset{\sim}{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ be a vector

$$\underset{\sim}{v} \cdot \underset{\sim}{v} = x \times x + y \times y = x^2 + y^2 = |\underset{\sim}{v}|^2$$

Alternative:

$$\underset{\sim}{v} \cdot \underset{\sim}{v} = |\underset{\sim}{v}| . |\underset{\sim}{v}| \cos\theta$$
$$= |\underset{\sim}{v}|^2 \cos\theta$$

where $\theta$ is the angle between and $\underset{\sim}{v}$ and $\underset{\sim}{v}$

$\therefore \quad \theta = 0 \quad$ and $\quad \cos\theta = 1$

$\therefore \quad \underset{\sim}{v} \cdot \underset{\sim}{v} = |\underset{\sim}{v}|^2$

***Sample answer:***

$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \underset{\sim}{a} + \underset{\sim}{b}$

$\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD} = -\underset{\sim}{a} + k\underset{\sim}{b}$

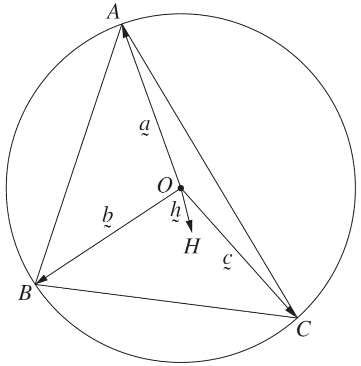
By part (i) $|\overrightarrow{AC}| = |\overrightarrow{BD}|$ implies

$$\overrightarrow{AC} \cdot \overrightarrow{AC} = \overrightarrow{BD} \cdot \overrightarrow{BD}$$
$$(\underset{\sim}{a} + \underset{\sim}{b}) \cdot (\underset{\sim}{a} + \underset{\sim}{b}) = (-\underset{\sim}{a} + k\underset{\sim}{b}) \cdot (-\underset{\sim}{a} + k\underset{\sim}{b})$$
$$\cancel{\underset{\sim}{a} \cdot \underset{\sim}{a}} + 2\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{b} = \cancel{\underset{\sim}{a} \cdot \underset{\sim}{a}} - 2k\underset{\sim}{a} \cdot \underset{\sim}{b} + k^2 \underset{\sim}{b} \cdot \underset{\sim}{b}$$
$$2(1+k)\underset{\sim}{a} \cdot \underset{\sim}{b} = (k^2 - 1)\underset{\sim}{b} \cdot \underset{\sim}{b}$$
$$2(1+k)\underset{\sim}{a} \cdot \underset{\sim}{b} = (k+1)(k-1)\underset{\sim}{b} \cdot \underset{\sim}{b}$$
$$1 + k \neq 0 \quad \text{since } k > 0$$
$$\therefore \quad 2\underset{\sim}{a} \cdot \underset{\sim}{b} = (k-1)\underset{\sim}{b} \cdot \underset{\sim}{b}$$
$$\therefore \quad 2\underset{\sim}{a} \cdot \underset{\sim}{b} = (k-1)|\underset{\sim}{b}|^2$$
$$2\underset{\sim}{a} \cdot \underset{\sim}{b} + (1-k)|\underset{\sim}{b}|^2 = 0$$

**2022 Q3**

Three different points $A$, $B$ and $C$ are chosen on a circle centred at $O$. **3**

Let $\underset{\sim}{a} = \overrightarrow{OA}$, $\underset{\sim}{b} = \overrightarrow{OB}$ and $\underset{\sim}{c} = \overrightarrow{OC}$. Let $\underset{\sim}{h} = \underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}$ and let $H$ be the point such that $\overrightarrow{OH} = \underset{\sim}{h}$, as shown in the diagram. Show that $\overrightarrow{BH}$ and $\overrightarrow{CA}$ are perpendicular.




We have $\overrightarrow{BH} = \underset{\sim}{h} - \underset{\sim}{b} = \underset{\sim}{a} + \underset{\sim}{c}$, while $\overrightarrow{CA} = \underset{\sim}{a} - \underset{\sim}{c}$. Hence

$$\begin{aligned}\overrightarrow{BH} \cdot \overrightarrow{CA} &= (\underset{\sim}{a} + \underset{\sim}{c}) \cdot (\underset{\sim}{a} - \underset{\sim}{c}) \\ &= \underset{\sim}{a} \cdot \underset{\sim}{a} - \underset{\sim}{a} \cdot \underset{\sim}{c} + \underset{\sim}{c} \cdot \underset{\sim}{a} - \underset{\sim}{c} \cdot \underset{\sim}{c} \\ &= \underset{\sim}{a} \cdot \underset{\sim}{a} - \underset{\sim}{c} \cdot \underset{\sim}{c} \\ &= |\underset{\sim}{a}|^2 - |\underset{\sim}{c}|^2 \\ &= 0 \quad \text{(as the lengths of } \underset{\sim}{a} \text{ and } \underset{\sim}{c} \text{ are equal, as they are radii of a circle)}\end{aligned}$$

Hence, the two vectors are perpendicular.

***Answers could include:***

We may observe that $\underset{\sim}{a} + \underset{\sim}{c}$ and $\underset{\sim}{a} - \underset{\sim}{c}$ form the diagonals of a rhombus, as $\underset{\sim}{a}$ and $\underset{\sim}{c}$ have equal length. The diagonals of a rhombus are perpendicular and so $\overrightarrow{BH}$ and $\overrightarrow{CA}$ are perpendicular.

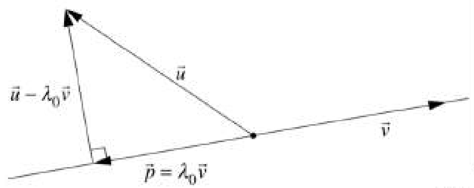
**2022 Q4**

The vectors $\vec{u}$ and $\vec{v}$ are not parallel. The vector $\vec{p}$ is the projection of $\vec{u}$ onto the vector $\vec{v}$. **3**

The vector $\vec{p}$ is parallel to $\vec{v}$ so it can be written $\lambda_0\vec{v}$ for some real number $\lambda_0$. (Do NOT prove this.)

Prove that $|\vec{u} - \lambda\vec{v}|$ is smallest when $\lambda = \lambda_0$ by showing that, for all real numbers $\lambda$, $|\vec{u} - \lambda_0\vec{v}| \le |\vec{u} - \lambda\vec{v}|$.

**METHOD 1**




Let $\lambda$ be a real number.

$$|\vec{u} - \lambda\vec{v}|^2 = |\vec{u} - \lambda_0\vec{v} + \lambda_0\vec{v} - \lambda\vec{v}|^2$$

$$= \Big|\underbrace{(\vec{u} - \lambda_0\vec{v})}_{\text{Perpendicular to } \vec{v}} + \underbrace{(\lambda_0 - \lambda)\vec{v}}_{\text{Parallel to } \vec{v}}\Big|^2$$

$$= \left[(\vec{u} - \lambda_0\vec{v}) + (\lambda_0 - \lambda)\vec{v}\right] \cdot \left[(\vec{u} - \lambda_0\vec{v}) + (\lambda_0 - \lambda)\vec{v}\right]$$

$$= |\vec{u} - \lambda_0\vec{v}|^2 + |(\lambda_0 - \lambda)\vec{v}|^2 + 0 + 0$$

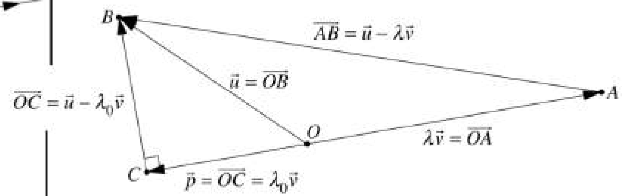
$$|(\lambda_0 - \lambda)\vec{v}|^2 \ge 0$$

Therefore $|\vec{u} - \lambda\vec{v}|^2 \ge |\vec{u} - \lambda_0\vec{v}|^2$.

So $|\vec{u} - \lambda\vec{v}| \ge |\vec{u} - \lambda_0\vec{v}|$.

Hence $|\vec{u} - \lambda\vec{v}|$ is smallest when it equals $|\vec{u} - \lambda_0\vec{v}|$, and so is smallest when $\lambda = \lambda_0$.

**METHOD 2**




Let $\lambda \in \mathbb{R}$.

Let $O$ be a point in the plane.

Let $A$, $B$ and $C$ be the points defined by

$\overrightarrow{OA} = \lambda\vec{v}$ $\quad \overrightarrow{OB} = \vec{u}$ $\quad \overrightarrow{OC} = \vec{p}$

Because $\vec{p}$ is the projection of $\vec{u}$ onto $\vec{v}$, the triangle $ABC$ has a right angle at $C$.

In any right-angled triangle the length of the hypotenuse is greater than or equal to the length of any of the sides

so $|\overrightarrow{AB}| \ge |\overrightarrow{BC}|$ $\quad$ That is $|\vec{u} - \lambda\vec{v}| \ge |\vec{u} - \lambda_0\vec{v}|$.

Hence $|\vec{u} - \lambda\vec{v}|$ is smallest when it equals $|\vec{u} - \lambda_0\vec{v}|$, and so is smallest when $\lambda = \lambda_0$.